

WASSIM CHOUAYAKH

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EDUCATION

National Institute of Applied Science and Technology (INSAT)

Tunis, Tunisia

Engineering Degree – Industrial IT and Automation (Expected 2028)

Sep 2025 – Present

National Institute of Applied Science and Technology (INSAT)

Tunis, Tunisia

Preparatory Cycle – Mathematics, Physics and Industrial IT

Sep 2023 – Jun 2025

TECHNICAL SKILLS

Languages: Python, Java, Dart, C/C++, JavaScript, TypeScript

Frameworks: FastAPI, React, Next.js, Flutter, LangGraph

AI / ML: PyTorch, TensorFlow, scikit-learn, MLflow, CNNs, NLP, RAG (LangChain, HuggingFace), LLM agents

DevOps / IaC: Git, Docker, Docker Compose, Terraform, Checkov, TFLint, CI/CD, Linux/Unix, AWS, Azure, GCP

Databases: PostgreSQL, MySQL, Redis

Other: REST APIs, WebSockets, Celery, Agile/Scrum

PROJECTS

Re-Tech Fusion – IoT Energy Monitoring Platform | *Python, C++, Docker, Streamlit, MQTT, ESP32* 2026

- ETL from Excel (52-measure cogeneration reports), PDFs (native + scanned via Gemini LLM) and scans (OpenCV + Tesseract OCR); unit-aware normalization across 31 physical units
- Docker Compose 4-service (Mosquitto, batch pipeline, ESP32 simulator, Streamlit); plugin-based ESP32 firmware (MQTT offline ring-buffer), 3-detector anomaly ensemble, CO₂ estimation

SightLine – AI Safety & Compliance Platform | *FastAPI, YOLOv8, HuggingFace, React, Docker, Python* 2025

- Real-time PPE violation detection at ~92% accuracy using YOLOv8 fine-tuned on 1,000+ images
- RAG engine (HuggingFace) over 50+ regulatory PDFs; Dockerized deployment cutting deployment time by ~60%
- React dashboard (live video feed, alerts, PDF compliance reports) for 5+ operators, cutting alert triage time by ~45%

RAG-Based Enterprise Knowledge Chatbot | *LangChain, HuggingFace, FastAPI, PostgreSQL, Docker* 2025

- Production RAG chatbot over 100+ internal documents; ~30% better response relevance and ~50% faster search vs. baseline LLM
- FAISS vector search + semantic chunking behind a FastAPI REST API, consumed by a React frontend; sub-200 ms query latency
- Docker Compose stack with re-indexing on document uploads; ~70% faster re-deployment, index freshness within 5 minutes

WORK EXPERIENCE

Information Systems Intern – PAMS

Sep 2026 – Present

UGFS North Africa – Tunis, Tunisia

- Contributing to the information system transformation program (PAMS), helping modernize core business processes
- Collaborating with business and IT teams on the deployment and adoption of the new system

AI Agent Developer – IaC Terraform Generator

Jul – Aug 2026

LOOYAS – Tunis, Tunisia

- LangGraph multi-agent pipeline (planner, contractor, coder, validator, security scanner, packager) generating Terraform for AWS, Azure and GCP from plain-text prompts
- Automated quality via terraform fmt/init/validate, TFLint, Checkov; deterministic HCL auto-fix + LLM refinement; stack: FastAPI, Celery, Redis, React, Docker Compose

Software Engineering Intern

Jul – Aug 2025

KPIT Technologies – Sfax, Tunisia

- Python test-case compliance automation cutting manual review time by ~40%; reports went from hours to minutes, saving ~3 h per review cycle
- Cross-platform desktop application (Flutter + Python backend) adopted by 10+ internal testers, reducing validation turnaround by ~30%

AI Engineering Intern

Jun – Jul 2025

Intellitech – Sfax, Tunisia

- RAG-based chatbot (LLM + retrieval over 100+ files); ~30% better relevance, ~4 h/week saved per analyst
- Automated ingestion (PDF, DOCX, TXT) via async FastAPI, 100+ documents per batch, zero manual intervention

LANGUAGES

French: Native **Arabic:** Native **English:** Fluent (professional) **Spanish:** Beginner